

Parameter Estimation of a Parametric Curve

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1 Problem Statement

The objective of this assignment is to estimate the three unknown parameters θ , M , and X of the following parametric curve:

$$x(t) = t \cos(\theta) - e^{M|t|} \sin(0.3t) \sin(\theta) + X, \quad (1)$$

$$y(t) = 42 + t \sin(\theta) + e^{M|t|} \sin(0.3t) \cos(\theta). \quad (2)$$

The available observations consist of 1500 (x, y) points stored in `xy_data.csv`. The parameter constraints are:

$$0^\circ < \theta < 50^\circ, \quad (3)$$

$$-0.05 < M < 0.05, \quad (4)$$

$$0 < X < 100, \quad (5)$$

with the parameter t satisfying

$$6 < t < 60.$$

The objective is to determine the values of θ , M , and X that best explain the observed data.

2 Mathematical Formulation

Define

$$f(t) = e^{M|t|} \sin(0.3t).$$

The parametric equations can then be written as

$$x(t) = t \cos(\theta) - f(t) \sin(\theta) + X, \quad (6)$$

$$y(t) = 42 + t \sin(\theta) + f(t) \cos(\theta). \quad (7)$$

Since the given range satisfies $t > 6$, we have

$$|t| = t.$$

Therefore,

$$f(t) = e^{Mt} \sin(0.3t).$$

For a candidate parameter vector

$$p = (\theta, M, X),$$

a predicted curve was generated by uniformly sampling t over the interval $[6, 60]$.

3 Parameter Estimation Method

The observed dataset contains 1500 two-dimensional points. For each candidate set of parameters (θ, M, X) , a predicted curve was generated using uniformly sampled values of t .

The distance between an observed point (x_i, y_i) and the predicted curve was measured using the L_1 distance:

$$d_1 = |x_i - x(t)| + |y_i - y(t)|.$$

For each observed point, the minimum distance to the sampled predicted curve was selected. The optimization objective was therefore

$$L(\theta, M, X) = \sum_{i=1}^N \min_t [|x_i - x(t)| + |y_i - y(t)|],$$

where $N = 1500$.

The parameter search was constrained to the ranges specified in the problem statement.

To perform the optimization, the `differential_evolution` algorithm from SciPy was used. This method performs a global search over the parameter space and was followed by local polishing to obtain a more accurate solution.

4 Implementation

The implementation was written in Python using NumPy, Pandas, SciPy, and Matplotlib.

The main steps were:

1. Load the 1500 observed (x, y) points.
2. Generate uniformly sampled values of t between 6 and 60.
3. Generate the predicted curve for candidate values of θ , M , and X .
4. Calculate the minimum L_1 distance between every observed point and the predicted curve.
5. Minimize the total distance using differential evolution.
6. Plot the observed points together with the fitted curve.

5 Results

The optimization produced the following parameter estimates:

Parameter	Estimated Value	Recovered Value
θ	29.99998244°	30°
M	0.03000003	0.03
X	54.99993411	55

Table 1: Estimated parameters of the parametric curve.

The resulting objective value was

$$L_1 = 6.109994.$$

The optimized parameters are extremely close to the simple values

$$\theta = 30^\circ, \quad M = 0.03, \quad X = 55.$$

The small difference between the numerical estimates and these values is attributable to numerical optimization and the discretization of the curve.

6 Recovered Curve

Substituting the recovered parameters gives the following parametric curve:

$$x(t) = t \cos(30^\circ) - e^{0.03|t|} \sin(0.3t) \sin(30^\circ) + 55, \quad (8)$$

$$y(t) = 42 + t \sin(30^\circ) + e^{0.03|t|} \sin(0.3t) \cos(30^\circ). \quad (9)$$

Since $t > 6$,

$$|t| = t,$$

so the final expression can also be written as

$$x(t) = t \cos(30^\circ) - e^{0.03t} \sin(0.3t) \sin(30^\circ) + 55, \quad (10)$$

$$y(t) = 42 + t \sin(30^\circ) + e^{0.03t} \sin(0.3t) \cos(30^\circ). \quad (11)$$

7 Validation

The fitted curve was plotted together with the 1500 observed data points. The fitted curve closely follows the structure of the observed point cloud.

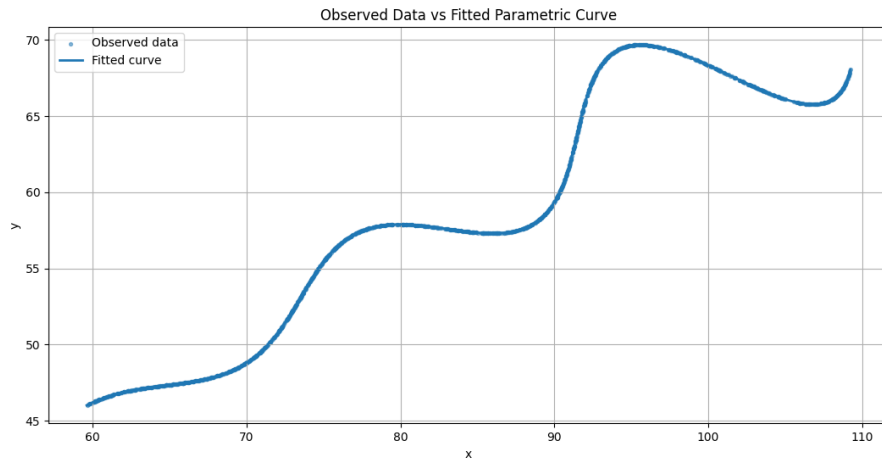


Figure 1: Observed data points and the fitted parametric curve.

The low total L_1 distance of 6.109994 provides quantitative evidence that the recovered parameters provide a close fit to the observed data.

8 Conclusion

The three unknown parameters of the given parametric curve were successfully estimated from the provided dataset.

The recovered values are:

$$\theta = 30^\circ, \quad M = 0.03, \quad X = 55$$

These values satisfy all the specified parameter bounds and produce a very small L_1 fitting error.

Thus, the recovered parametric curve is consistent with the provided (x, y) observations.

9 Code

The complete Python implementation used for parameter estimation is provided separately in the project repository.

The implementation uses:

- NumPy for numerical computation,
- Pandas for reading the CSV dataset,
- SciPy for global optimization,
- Matplotlib for visualization.

GitHub repository:

https://github.com/saniyaahemad12/Parametric_Curve_Parameter_Estimation